Convolution Neural Network

Agenda

- Quick Neural Network Recap
- Introduction to CNN
- Convolution Operation
- Kernel/Filter Design and Interpretation
- Pooling layer
- Fully Connected layer in CNN
- Batch Normalization & Regularization
- Training and Optimization(Data Augmentation)

Quick Recap Neural Network

- The Convolution Neural Network(ConvNet), is a deep neural network specialized for Image Recognition.
- It demonstrate how significant the improvement of the deep layers is for information(image) processing.
- Image Recognition is classification task:
  - To demonstrate is the image belongs to cat or dog.
- The output layer of ConvNet generally employees the multiclass classification neural network.

Convolution Neural Network

- An image of size 600 x 600 pixels with three color channels will have 1,080,000 pieces of information encoded (600 x 600 x 3)
- Therefore our input layer would require 1,080,000 neurons.
- If the next layer in the network contains 1,000 neurons, we'd need to maintain one billion weights between the first two layers alone.
- Dimensionality reduction can not solve this problem, the network will still be large.
- The is where CNN comes to play.

Convolution Neural Network

- In short, a CNN is an ANN with multiple-perhaps many- preprocessing layers that perform transformation on the image before ultimately before reaching to final fully connected layer.
- By incorporating the preprocessing tasks into the network, the back propagation algorithm can tune the preprocessing tasks as a part of network training.
- The network will not only learn how to classify image, it also learn how to preprocess images to your task.

Convolution Neural Network

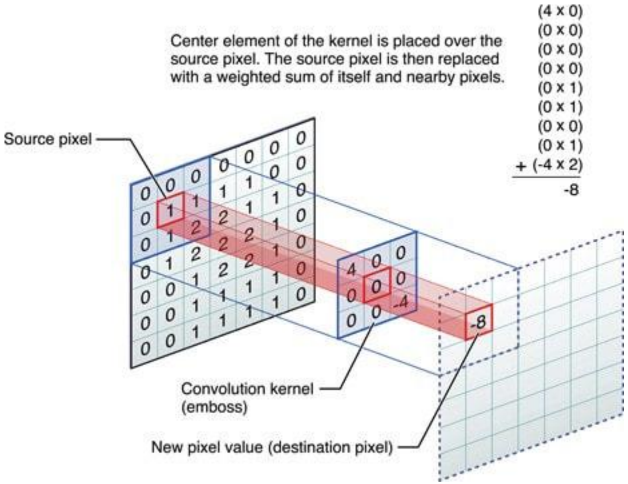
- The feature extraction neural network consists of piles of the Convolution Layer and Pooling layer pair.
- The convolution layer converts the image using the convolution operation.
    - It can be thought of as a collection of digital filters.
- The pooling layer combines the neighboring pixels into a single pixel-therefore, it reduces the dimension of the image.

Convolution Neural Network

- In an image context, convolutions are represented by a convolution matrix, which is typically a small matrix(3*3, 5*5, or similar).
- The convolution matrix is much smaller than image to be proceed.
- The convolution matrix, is swept across the image so the output of the convolution applied to the entire image builds a new image with effect applied.

Convolution Neural Network-Filter Matrix

- Here is how the convolution of an image with a convolution filter works:



- Each filter of the first (leftmost) layer slides, across the input image, left to right, top to bottom, and convolution is computed at each iteration.

Convolution Neural Network

- The filter matrix (one for each filter in each layer) and bias values are re trainable parameters that are optimized using gradient descent with back propagation.
- A nonlinearity is applied to the sum of the convolution and the bias term.
- Typically, the RELU activation function is used in all hidden layers.
- The activation function of the output layer depends on the task.

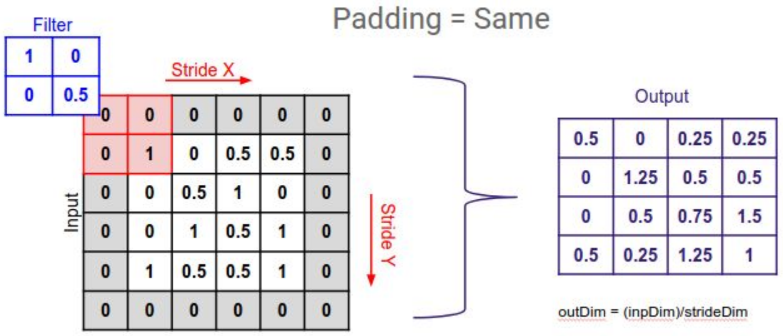
Stride and Padding

- Two important properties of convolution are stride and padding.
- It has affect how the convolutional filter(kernel) slides over your input(e.g., an image) and how large the resulting feature map will be.

Stride

- Definition: Stride is the amount by which the filter “slides”(move) across the image or feature map at each step.
- Common values: Often set to 1(moves by 1 pixel each time) or 2(moves by 2 pixel)
- Effect of output:
  - A larger stride(e.g., 2) produce the smaller output feature map because the filter covers the input fewer positions.
  - A smaller stride (e.g.,1) produces a larger output because the filter scans the input more closely.

Padding

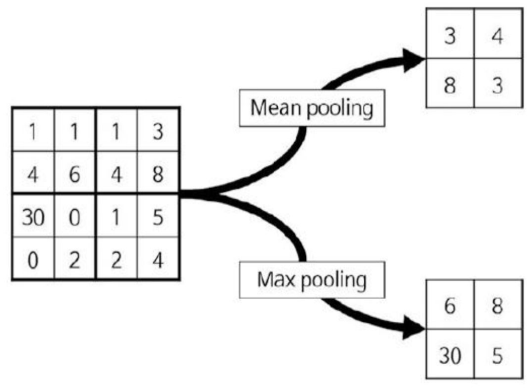
- Definition:
  - Padding adds extra pixels (usually zeros, called zero-padding) around the border of the input image before applying convolution.
- Purpose:
  - Control the output size: you can preserve the same spatial dimensions after convolution if you pad the input appropriately (often called same padding)
  - 

Convolution Layer- Feature map

- The features that the convolution layer extracts vary depending on which convolution filter is used.
- The feature map that convolution filter creates is processed through the activation function before the layer yields the output.
    - Relu function is used in most of the recent applications, the sigmoid function and the tanh function are often employed as well.

Pooling layer

- The pooling layer reduces the size of the image, as it combines neighboring pixels of a certain area of the image into a single representative value.
- The figure below shows the resultant cases of pooling using the mean pooling and max pooling.

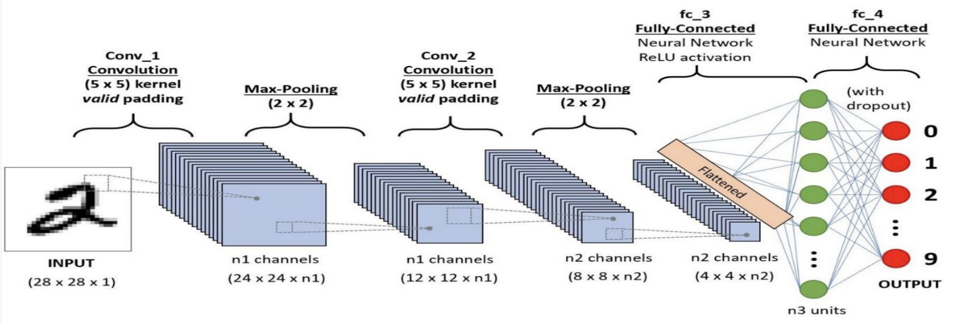


(1+3+4+8)/4 = 4
(30+0+0+2)/4 = 8
(1+5+2+4)/4 = 3

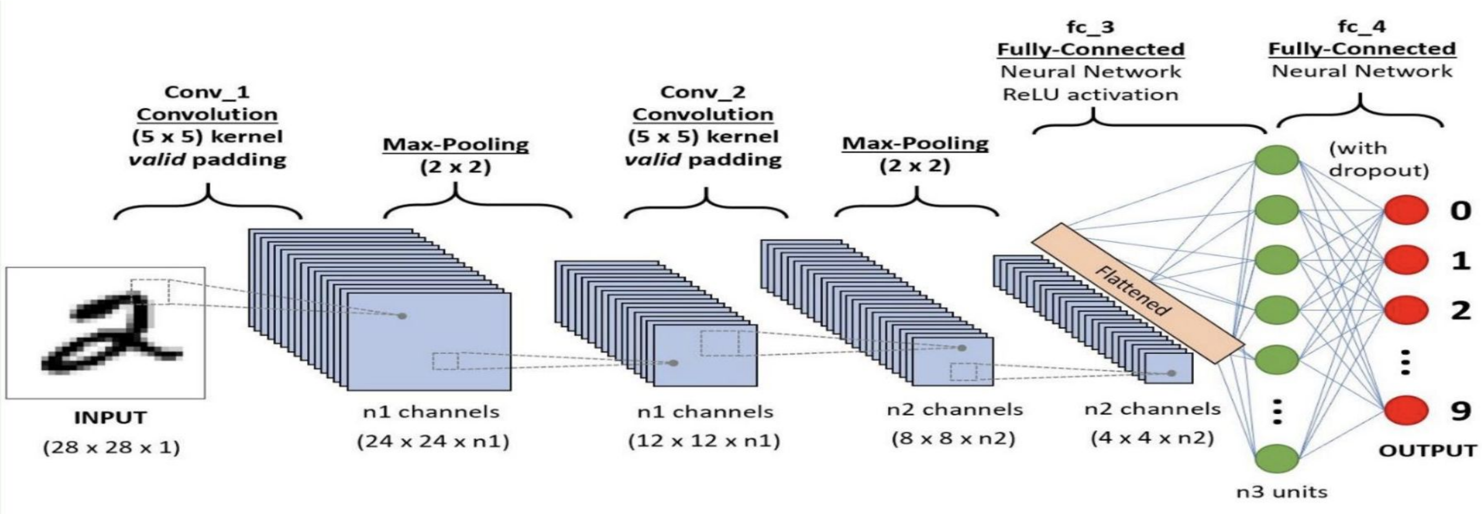
max (1,1,4,6) = 6
max (1,3,4,8) = 8
max (30,0,0,2) = 30
max (1,5,2,4) = 5

Using CNN

- A CNN uses multiple convolving layers, each with multiple convolution filters, to build a model of the image.
- The convolution layers and the convolution filters themselves are trained by the backpropagation algorithm, and the network will eventually discover the correct filters to use in order to enhance the features that the network is trying to identify.

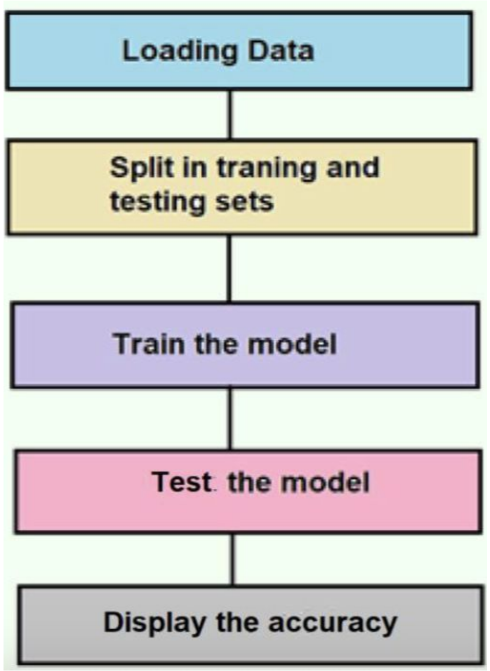


Using CNN



Conv_1
Convolution
(5 x 5) kernel
valid padding
Max-Pooling
(2 x 2)
Conv_2
Convolution
(5 x 5) kernel
valid padding
Max-Pooling
(2 x 2)
fc_3
Fully-Connected
Neural Network
ReLU activation
fc_4
Fully-Connected
Neural Network
(with dropout)
Flattened
INPUT
(28 x 28 x 1)
n1 channels
(24 x 24 x n1)
n1 channels
(12 x 12 x n1)
n2 channels
(8 x 8 x n2)
n2 channels
(4 x 4 x n2)
n3 units
0
1
2
9
OUTPUT

Process of Computer vision tasks



Questions?